Microsoft

PostgreSQL and GraphRAG Integration in Docker

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Microsoft

Background and objectives:

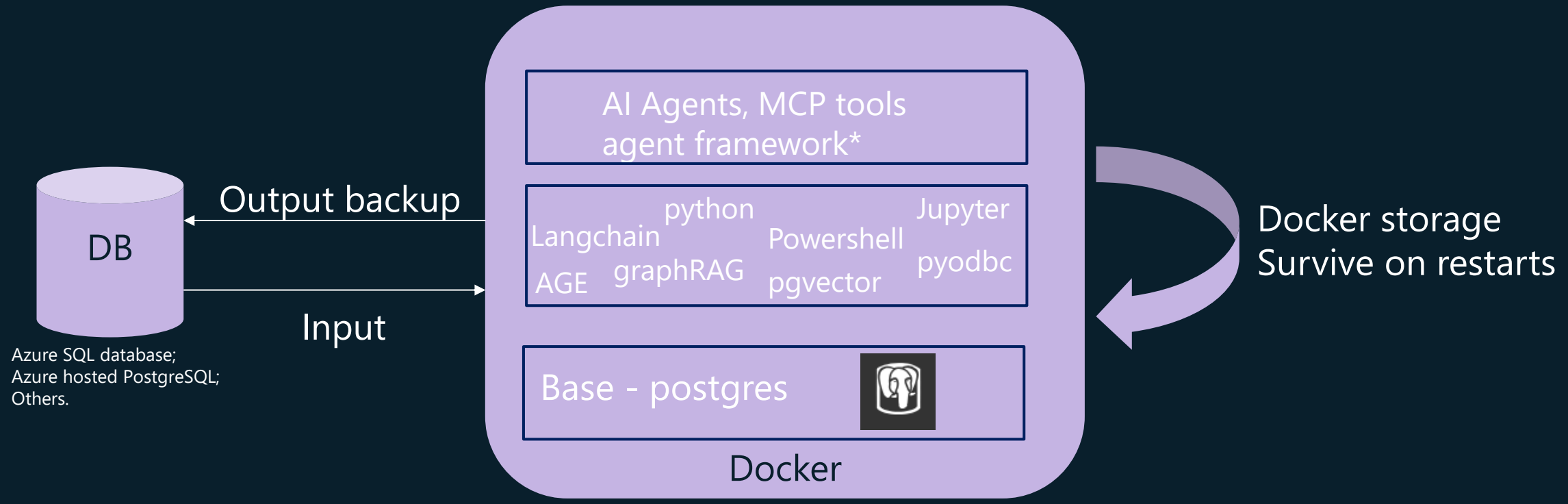
- Raw input data is already in a DB.
- Open Cypher query capability is needed.
- Run CLI version of graphRAG Conveniently.
- Avoid another blob storage between the DB and graphRAG.

Solution design:

- Build graphARG index directly using data in the DB.
- Integrate PostgreSQL and graphRAG in a Docker container, add Apache AGE for running Cypher query.
- Create agents and include MCP tools.
- Agents are equipped with capability of converting natural language to Cypher query.

Architecture

The docker image uses postgres as base, then added python, graphRAG, AGE, semantic-kernel and other needed packages. Two volumes are created to persist postgres data and app related data.




```yaml
volumes:
  - local_postgres_data:/var/lib/postgresql/data
  - graphrag_data:/app/graphrag-folder
```

MCP Tools

[graphrag_search]:
description="Run a GraphRAG query (local or global) with runtime-tunable API params",

[age_get_schema_cached]:
description="Return cached AGE graph schema; if refresh=true, re-query the database and update the cache."

[age_cypher_query]:
description="Execute a user-provided Cypher query against the AGE graph and return rows (each row under key 'result')."

[age_entity_lookup]:
description="Find Entity nodes by name/title substring match (best for 'Who is X?' or quick disambiguation)."

[age_nl2cypher_query]
description="Convert a natural-language question into a Cypher query (Entity/RELATED_TO only), execute it, and return rows; best for complex or multi-hop semantic graph questions."

Preparation

1. Prepare Dockerfile
2. Prepare the .env file
3. Prepare setting.yaml
4. Prepare docker-compose.yaml
5. Prepare python code or jupyter notebook

.env example

Your DB type,
you can define yours

Your DB properties

Your docker Postgres property

If you use your docker
Postgres, then set to true

```
DB_TYPE="Azure SQL"  # "Azure SQL" or "Azure postgres"
MY_DB_HOST=
MY_DB_PORT=
MY_DB_USER=
MY_DB_PASSWORD=
MY_DB_NAME=
MY_DB_SSLMODE=disable
PROMPTS=/app/graphrag-folder/prompts
CONFIG_PATH=/app/graphrag-folder/settings.yaml
AGE_HOST=postgres
AGE_PORT=5432
AGE_USER=postgres
AGE_PASSWORD=xxxx
AGE_DB=postgres
USE_LOCAL_AGE=true
AOAI_API_BASE="https://xxxx.openai.azure.com"
AOAI_API_KEY=xxxx
AOAI_LLM_MODEL=gpt-4o
AOAI_LLM_API_VERSION="2025-01-01-preview"
AOAI_LLM_DEPLOYMENT=gpt-4o
AOAI_EMBEDDING_MODEL=text-embedding-3-small
AOAI_EMBEDDING_API_VERSION="2023-05-15"
AOAI_EMBEDDING_DEPLOYMENT=text-embedding-3-small
```

Step 0 – insert .txt input to DB if not already existing

Run python code to insert .txt files to DB:

➢ python insert-table.py

The code insert-table.py provides two methods of DB connections:

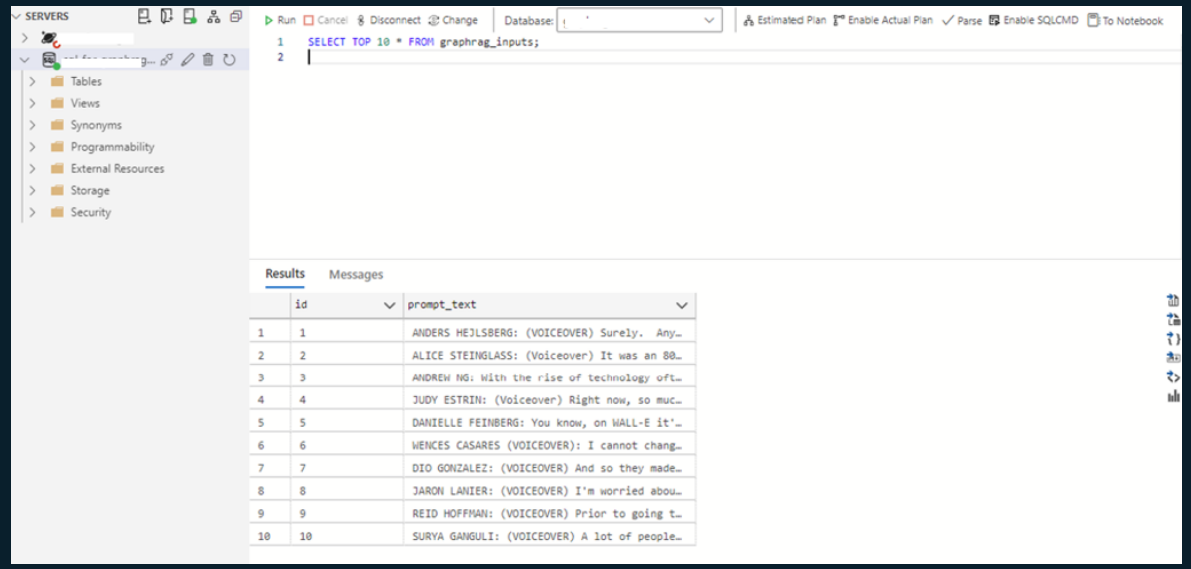
- For inserting table to Azure hosted PostgreSQL.
- Or for inserting table to Azure SQL database.

Below is example of inserting to Azure SQL database:

> Python insert-table.py

```
['SQL Server', 'Microsoft Access Driver (*.mdb, *.accdb)', 'Microsoft Excel Driver (*.xls, *.xlsx, *.xlsm, *.xlsb)', 'Microsoft Access Text Driver (*.txt, *.csv)',
'Microsoft Access dBASE Driver (*.dbf, *.ndx, *.mdx)', 'ODBC Driver 18 for SQL Server']
Processed file: EP-01_Behind_the_Tech_Kevin_Scott_Podcast_with_Anders_Hejlsberg.txt
Number of lines in EP-01_Behind_the_Tech_Kevin_Scott_Podcast_with_Anders_Hejlsberg.txt: 241
Processed file: EP-02_Behind_the_Tech_Kevin_Scott_Podcast_with_Alice_Steinglass.txt
Number of lines in EP-02_Behind_the_Tech_Kevin_Scott_Podcast_with_Alice_Steinglass.txt: 207
Processed file: EP-03_Behind_the_Tech_Kevin_Scott_Podcast_with_Andrew_Ng.txt
Number of lines in EP-03_Behind_the_Tech_Kevin_Scott_Podcast_with_Andrew_Ng.txt: 159
Processed file: EP-04_Behind_the_Tech_Kevin_Scott_Podcast_with_Judy_Estrin.txt
Number of lines in EP-04_Behind_the_Tech_Kevin_Scott_Podcast_with_Judy_Estrin.txt: 200
Processed file: EP-05_Behind_the_Tech_Kevin_Scott_Podcast_with_Danielle_Feinberg.txt
Number of lines in EP-05_Behind_the_Tech_Kevin_Scott_Podcast_with_Danielle_Feinberg.txt: 291
Processed file: EP-06_Behind_the_Tech_Kevin_Scott_Podcast_with_Wences_Casares.txt
Number of lines in EP-06_Behind_the_Tech_Kevin_Scott_Podcast_with_Wences_Casares.txt: 265
Processed file: EP-07_Behind_the_Tech_Kevin_Scott_Podcast_with_Dio_Gonzalez.txt
Number of lines in EP-07_Behind_the_Tech_Kevin_Scott_Podcast_with_Dio_Gonzalez.txt: 702
Processed file: EP-08_Behind_the_Tech_Kevin_Scott_Podcast_with_Jason_Lanier.txt
Number of lines in EP-08_Behind_the_Tech_Kevin_Scott_Podcast_with_Jason_Lanier.txt: 426
Processed file: EP-09_Behind_the_Tech_Kevin_Scott_Podcast_with_Reid_Hoffman.txt
Number of lines in EP-09_Behind_the_Tech_Kevin_Scott_Podcast_with_Reid_Hoffman.txt: 388
Processed file: EP-10_Behind_the_Tech_Kevin_Scott_Podcast_with_Surya_Ganguli.txt
Number of lines in EP-10_Behind_the_Tech_Kevin_Scott_Podcast_with_Surya_Ganguli.txt: 254
All .txt files inserted into Azure SQL Database.
```

Azure Data Studio

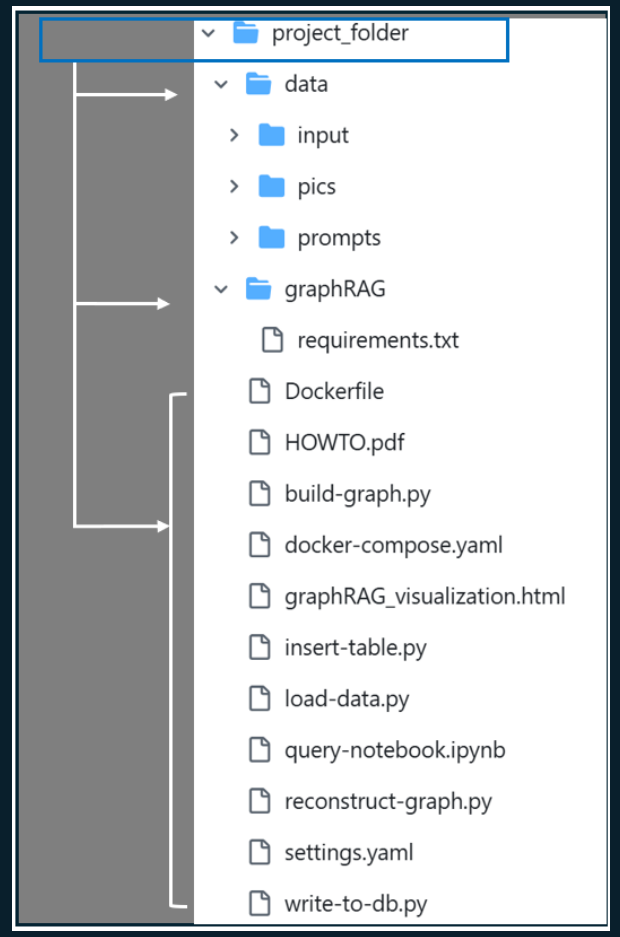


Step 1 – build docker image

This step builds docker images to include all modules needed to run services later.

In project_folder:

```
$ docker build -t graphrag-img .
```



Step 2 – run postgres service

This step starts postgres service which other services depend on.

```
$ docker compose up postgres
```

```
[+] Running 2/2
 ✓ Network postgresql-age_default  Created                                                    0.1s
 ✓ Container postgres              Created                                                    0.1s
Attaching to postgres
postgres  |
postgres  | PostgreSQL Database directory appears to contain a database; Skipping initialization
postgres  |
postgres  | 2025-05-23 16:28:27.226 UTC [1] LOG:  starting PostgreSQL 16.9 (Debian 16.9-1.pgdg120+1) on x86_64-pc-linux-gnu, compil
ed by gcc (Debian 12.2.0-14) 12.2.0, 64-bit
postgres  | 2025-05-23 16:28:27.227 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
postgres  | 2025-05-23 16:28:27.227 UTC [1] LOG:  listening on IPv6 address "::", port 5432
postgres  | 2025-05-23 16:28:27.234 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
postgres  | 2025-05-23 16:28:27.250 UTC [29] LOG:  database system was shut down at 2025-05-23 16:05:41 UTC
postgres  | 2025-05-23 16:28:27.270 UTC [1] LOG:  database system is ready to accept connections
```

Step 3.1 – run ‘load-data’ service

This step loads data from DB to Docker folder /app/graphrag-folder/input as graphRAG input.

```
$ docker compose up load-data
```

```
[+] Running 2/2
 ✓ Container postgres       Running                                                    0.0s
 ✓ Container load-data-app  Created                                                    0.1s
Attaching to load-data-app
load-data-app  | All rows saved as .txt files in /app/graphrag-folder/input.
load-data-app exited with code 0
```

Step 3.2 – check data is present in postgres server

This sub-step checks docker folder /app/graphrag-folder/input to verify graphRAG input.

```
$ docker exec -it postgres bash
```

```
root@fffff:/app/graphrag-folder# cd input
root@fffff:/app/graphrag-folder/input# ls
input_10.txt  input_1.txt  input_2.txt  input_3.txt  input_4.txt  input_5.txt  input_6.txt  input_7.txt  input_8.txt  input_9.txt
```

Step 4.1 – build graphrag index

This step runs ‘graphrag index’ using /app/graphrag-folder/input and generate output.

```
$ docker compose up graphrag-index
```

```
[+] Running 3/3
 ✓ Container postgres            Running                                                    0.0s
 ✓ Container load-data-app       Created                                                    0.0s
 ✓ Container graphrag-index-app  Created                                                    0.1s
Attaching to graphrag-index-app
graphrag-index-app  | Total files: 10
```

```
graphrag-index-app  |
graphrag-index-app  | [119 rows x 2 columns]}
graphrag-index-app  | ⠋ GraphRAG Indexer
graphrag-index-app  | ├── Loading Input (InputFileType.text) - 10 files loaded (0 filtered) ━ 100%  0…
graphrag-index-app  | ├── create_base_text_units ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── create_final_documents ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── extract_graph ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── finalize_graph ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── create_communities ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── create_final_text_units ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── create_community_reports ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── generate_text_embeddings ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00🚀 All workflows completed su
ccessfully.
graphrag-index-app exited with code 0
```

Step 4.2 – check data is present in postgres server

This sub-step verifies ‘graphrag index’ output in /app/graphrag-folder/output.

```
$ docker exec -it postgres bash
```

```
root@fffff:/app/graphrag-folder# cd output
root@fffff:/app/graphrag-folder/output# ls
communities.parquet        embeddings.community.full_content.parquet  graph.graphml          text_units.parquet
community_reports.parquet  embeddings.entity.description.parquet      lancedb
context.json               embeddings.text_unit.text.parquet          relationships.parquet
documents.parquet          entities.parquet                           stats.json
root@fffff:/app/graphrag-folder/output# cd lancedb
root@fffff:/app/graphrag-folder/output/lancedb# ls
default-community-full_content.lance  default-entity-description.lance  default-text_unit-text.lance
```

Step 5 – write index output to DB

This step stores the graph output data in DB, as docker backup.
The content in /app/graphrag-folder/output will be saved into DB. This example uses Azure SQL database as DB.

```
$ docker compose up graphrag-writer
```

```
[+] Running 4/4
 ✓ Container postgres             Running                                                    0.0s
 ✓ Container load-data-app        Created                                                    0.0s
 ✓ Container graphrag-index-app   Created                                                    0.0s
 ✓ Container graphrag-writer-app  Recreated                                                  0.2s
Attaching to graphrag-writer-app
graphrag-writer-app  | All relevant files have been processed and stored in Azure SQL .
graphrag-writer-app exited with code 0
```

Step 6 – build graph, prepare for Cypher query

This step builds graph using AGE on PostgreSQL in docker, to prepare for Cypher query.

```
$ docker compose up build-graph
```

```
[+] Running 5/5
 ✓ Container postgres            Running                                    0.0s
 ✓ Container load-data-app       Created                                    0.0s
 ✓ Container graphrag-index-app  Created                                    0.0s
 ✓ Container graphrag-writer-app Created                                    0.0s
 ✓ Container build-graph-app     Recreated                                  0.1s
Attaching to build-graph-app
build-graph-app  | Existing graph 'graphRAG' dropped.
build-graph-app  | Graph 'graphRAG' will be created.
build-graph-app  | Inserted 10 rows into Document
build-graph-app  | Index(['id', 'human_readable_id', 'title', 'text', 'text_unit_ids',
build-graph-app  |        'creation_date', 'metadata'],
build-graph-app  |       dtype='object')
build-graph-app  | Inserted 437 rows into Entity
build-graph-app  | Index(['id', 'human_readable_id', 'title', 'type', 'description',
build-graph-app  |        'text_unit_ids', 'frequency', 'degree', 'x', 'y'],
build-graph-app  |       dtype='object')
build-graph-app  | Inserted 696 rows into relationship
build-graph-app  | Index(['id', 'human_readable_id', 'source', 'target', 'description', 'weight',
build-graph-app  |        'combined_degree', 'text_unit_ids'],
build-graph-app  |       dtype='object')
build-graph-app  | Data loaded into AGE graph successfully.
build-graph-app exited with code 0
```

View the graph content

This step verifies graph in PostgreSQL ag_catalog in docker, name graphRAG.

```
$ docker exec -it postgres bash
```

```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()--() RETURN count(*)
$$) AS (count agtype);
 count
-------
 1392
(1 row)

postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()-[r]->() RETURN count(r)
$$) AS (count agtype);
 count
-------
 696
(1 row)
```

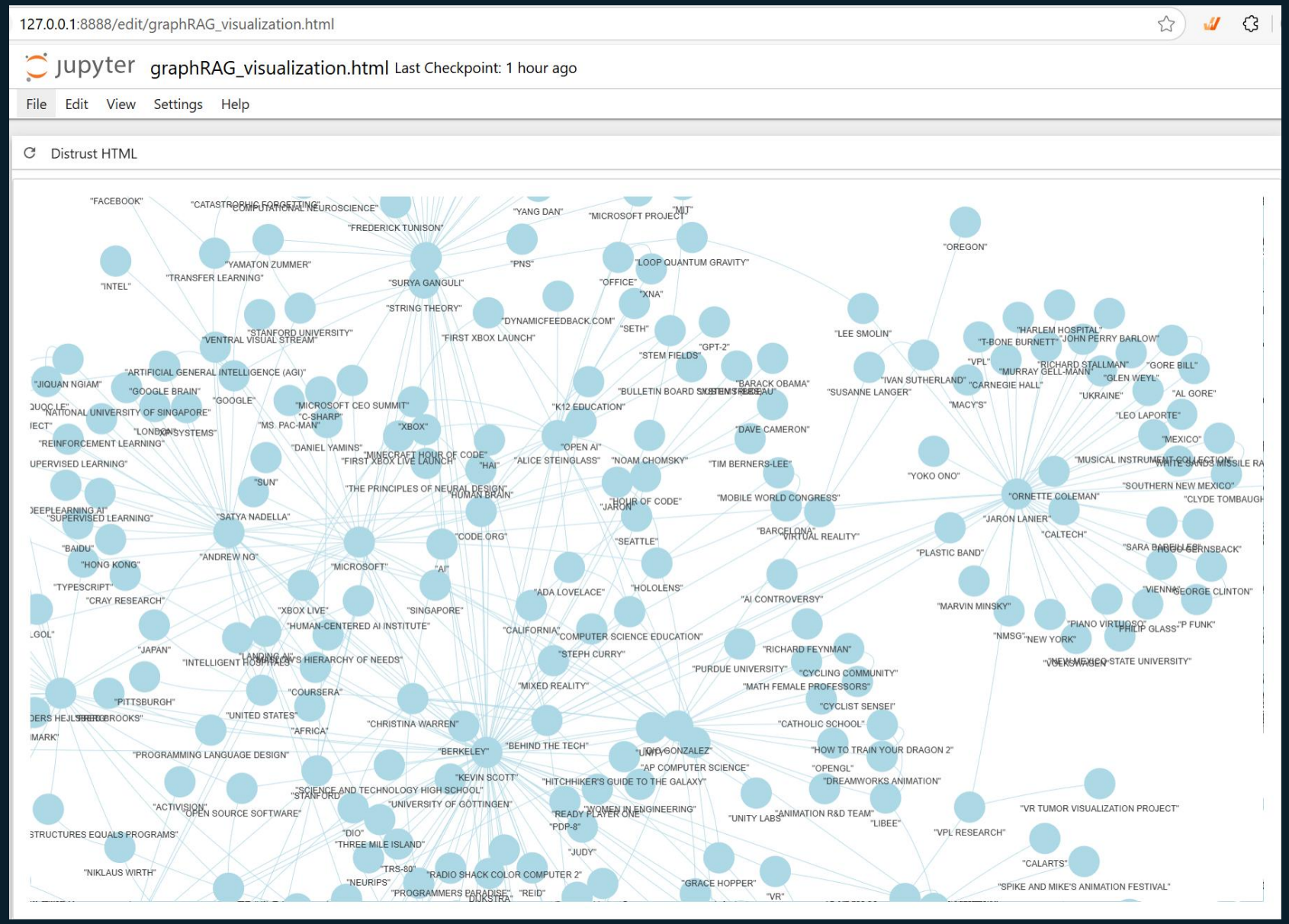
```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# \dt
              List of relations
   Schema   |   Name   | Type  |  Owner
------------+----------+-------+----------
 ag_catalog | ag_graph | table | postgres
 ag_catalog | ag_label | table | postgres
(2 rows)

postgres=# SELECT * FROM ag_catalog.ag_label;
       name        | graph | id | kind |          relation           |        seq_name
------------------+-------+----+------+-----------------------------+------------------------
 _ag_label_vertex | 87167 |  1 | v    | "graphRAG"._ag_label_vertex | _ag_label_vertex_id_seq
 _ag_label_edge   | 87167 |  2 | e    | "graphRAG"._ag_label_edge   | _ag_label_edge_id_seq
 Document         | 87167 |  3 | v    | "graphRAG"."Document"       | Document_id_seq
 Entity           | 87167 |  4 | v    | "graphRAG"."Entity"         | Entity_id_seq
 RELATED_TO       | 87167 |  5 | e    | "graphRAG"."RELATED_TO"     | RELATED_TO_id_seq
(5 rows)


postgres=# SELECT * FROM ag_catalog.ag_graph;
 graphid |   name   | namespace
---------+----------+------------
   87167 | graphRAG | "graphRAG"
(1 row)
```

Graph visualization



Step 7 – run query in jupyter notebook

This step runs jupyter notebook in docker.

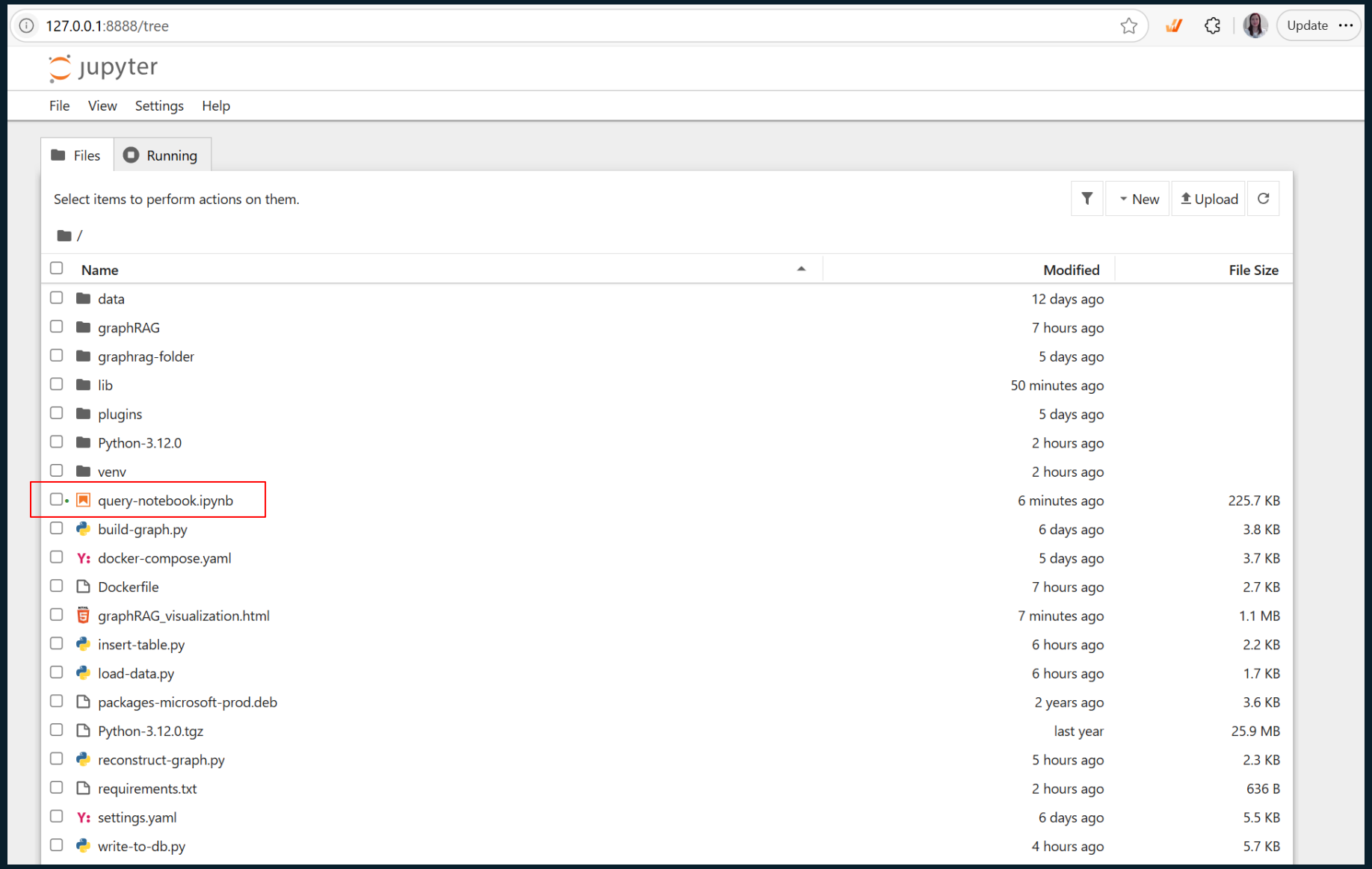
```
$ docker compose up query-notebook
```

```
[+] Running 6/6
 ✓ Container postgres            Running                                    0.0s
 ✓ Container load-data-app       Created                                    0.0s
 ✓ Container graphrag-index-app  Created                                    0.0s
 ✓ Container graphrag-writer-app Created                                    0.0s
 ✓ Container build-graph-app     Created                                    0.0s
 ✓ Container query-notebook-app  Recreated                                  0.1s
Attaching to query-notebook-app
query-notebook-app  | [I 2025-05-23 16:51:30.721 ServerApp] jupyter_lsp | extension was successfully linked.
query-notebook-app  | [I 2025-05-23 16:51:30.727 ServerApp] jupyter_server_terminals | extension was successfully linked.
query-notebook-app  | [W 2025-05-23 16:51:30.728 LabApp] 'shutdown_no_activity_timeout' has moved from NotebookApp to ServerApp. Th
is config will be passed to ServerApp. Be sure to update your config before our next release.
query-notebook-app  | [I 2025-05-23 16:51:30.730 ServerApp] jupyterlab | extension was successfully linked.
query-notebook-app  | [I 2025-05-23 16:51:30.734 ServerApp] notebook | extension was successfully linked.
query-notebook-app  | [I 2025-05-23 16:51:30.742 ServerApp] Writing Jupyter server cookie secret to /root/.local/share/jupyter/runt
ime/jupyter_cookie_secret
query-notebook-app  | [I 2025-05-23 16:51:31.018 ServerApp] notebook_shim | extension was successfully linked.
query-notebook-app  | [I 2025-05-23 16:51:31.060 ServerApp] notebook_shim | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.062 ServerApp] jupyter_lsp | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.063 ServerApp] jupyter_server_terminals | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.078 LabApp] JupyterLab extension loaded from /app/venv/lib/python3.12/site-packages/ju
pyterlab
query-notebook-app  | [I 2025-05-23 16:51:31.078 LabApp] JupyterLab application directory is /app/venv/share/jupyter/lab
query-notebook-app  | [I 2025-05-23 16:51:31.080 LabApp] Extension Manager is 'pypi'.
query-notebook-app  | [I 2025-05-23 16:51:31.136 ServerApp] jupyterlab | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.138 ServerApp] notebook | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp] Serving notebooks from local directory: /app
query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp] Jupyter Server 2.16.0 is running at:
query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp] http://

query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp]     http://127.0.0.1:8888/tree?token=

query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to s
kip confirmation).
query-notebook-app  | [C 2025-05-23 16:51:31.150 ServerApp]
```

Jupyter notebook UI



Cypher query examples

Several ways of querying are provided:

- Cypher query
  - Example given to check nodes and relationships, and their properties.
- Multi-hop cypher query
  - One of the advantages of using knowledge graph is the multi-hop query.
- GraphRAG LocalSearch
  - Examples given to show the impact of community level to the query results.
- GraphRAG GlobalSearch
  - Both python function call and CLI methods are provided.
- Vector search
  - Example shows that vector search result is restricted to the original text; it typically returns the top-k most similar original chunks. In other words, vector search is retrieval only, it returns what was said, not what it means. However, graphRAG combines vector search with a knowledge graph that encodes relationships between entities. The graph is built from LLM-extracted triples, which helps contextual thinking; thus, graphRAG provides more synthesized and thoughtful response.

Two vector search methods

Use AOAI in local postgres, without pgvector

Use pgvector extension in local postgres.

It creates a table graphrag_vectors in postgres

```
postgres=# \dt
                List of relations
   Schema   |       Name       | Type  |  Owner
------------+------------------+-------+----------
 ag_catalog | ag_graph         | table | postgres
 ag_catalog | ag_label         | table | postgres
 public     | graphrag_vectors | table | postgres
(3 rows)

postgres=#
SELECT COUNT(*) FROM graphrag_vectors;
 count
-------
   119
(1 row)

postgres=#
SELECT * FROM graphrag_vectors LIMIT 5;
postgres=# \dx
                              List of installed extensions
  Name   | Version |   Schema   |                     Description
---------+---------+------------+-----------------------------------------------------
 age     | 1.5.0   | ag_catalog | AGE database extension
 plpgsql | 1.0     | pg_catalog | PL/pgSQL procedural language
 vector  | 0.8.0   | public     | vector data type and ivfflat and hnsw access methods
(3 rows)
```

Step 8 – run agent and MCP tools in jupyter notebook

This step runs jupyter notebook in docker.

```
$ docker compose up mcp-agent
```

```
[+] Running 6/6
 ✓ Container postgres              Running
                   0.0s
 ✓ Container load-data-app         Created
                   0.0s
 ✓ Container graphrag-index-app    Running
                   0.0s
 ✓ Container graphrag-writer-app   Running
                   0.0s
 ✓ Container build-graph-app       Created
                   0.0s
 ✓ Container mcp-agent-app         Created
                   0.0s
Attaching to mcp-agent-app
mcp-agent-app  | [I 2026-03-13 04:54:08.981 ServerApp] jupyter_lsp | extension was successfully linked.
mcp-agent-app  | [I 2026-03-13 04:54:08.991 ServerApp] jupyter_server_terminals | extension was successfully linked.
mcp-agent-app  | [W 2026-03-13 04:54:08.996 LabApp] 'shutdown_no_activity_timeout' has moved from NotebookApp to ServerApp. This config will be pa
ssed to ServerApp. Be sure to update your config before our next release.

mcp-agent-app  | [I 2026-03-13 04:54:09.006 ServerApp] jupyterlab | extension was successfully linked.
mcp-agent-app  | [I 2026-03-13 04:54:09.017 ServerApp] notebook | extension was successfully linked.
mcp-agent-app  | [I 2026-03-13 04:54:09.522 ServerApp] notebook_shim | extension was successfully linked.

mcp-agent-app  | [I 2026-03-13 04:54:09.559 ServerApp] notebook_shim | extension was successfully loaded.
mcp-agent-app  | [I 2026-03-13 04:54:09.564 ServerApp] jupyter_lsp | extension was successfully loaded.
mcp-agent-app  | [I 2026-03-13 04:54:09.566 ServerApp] jupyter_server_terminals | extension was successfully loaded.
mcp-agent-app  | [I 2026-03-13 04:54:09.577 LabApp] JupyterLab extension loaded from /opt/venv/lib/python3.12/site-packages/jupyterlab

mcp-agent-app  | [I 2026-03-13 04:54:09.577 LabApp] JupyterLab application directory is /opt/venv/share/jupyter/lab
mcp-agent-app  | [I 2026-03-13 04:54:09.578 LabApp] Extension Manager is 'pypi'.

mcp-agent-app  | [I 2026-03-13 04:54:09.699 ServerApp] jupyterlab | extension was successfully loaded.

mcp-agent-app  | [I 2026-03-13 04:54:09.707 ServerApp] notebook | extension was successfully loaded.
mcp-agent-app  | [I 2026-03-13 04:54:09.709 ServerApp] Serving notebooks from local directory: /app

mcp-agent-app  | [I 2026-03-13 04:54:09.710 ServerApp] Jupyter Server 2.17.0 is running at:

mcp-agent-app  | [I 2026-03-13 04:54:09.710 ServerApp] http://6d2bb80306cb:8084/tree?token=3cac5a84d80eeb9142ffa494b46dc57a8f46f8cb2b7d9ebb
mcp-agent-app  | [I 2026-03-13 04:54:09.711 ServerApp]     http://127.0.0.1:8084/tree?token=3cac5a84d80eeb9142ffa494b46dc57a8f46f8cb2b7d9ebb

mcp-agent-app  | [I 2026-03-13 04:54:09.713 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).

mcp-agent-app  | [C 2026-03-13 04:54:09.725 ServerApp]

mcp-agent-app  |
mcp-agent-app  |     To access the server, open this file in a browser:

mcp-agent-app  |         file:///root/.local/share/jupyter/runtime/jpserver-1-open.html

mcp-agent-app  |     Or copy and paste one of these URLs:

mcp-agent-app  |         http://6d2bb80306cb:8084/tree?t
mcp-agent-app  |         http://127.0.0.1:8084/tree?t
```

Access agent-notebook.ipynb

Agent scenarions Included

Multiple scenarions of agents with MCP tools are included in the *agent-notebook.ipynb*:

- graphRAG search: local search and global search examples with direct mcp call.
- graphRAG search: local search and global search examples with agent and include mcp tools.
- Cypher query in direct mcp call.
- Agent to query in natural language, and mcp tool included to convert NL2Cypher.
- Agent with unified mcp (all five mcp tools), and based on the question route to the corresponding tool.
  - ['graphrag_search', 'age_get_schema_cached', 'age_cypher_query', 'age_entity_lookup', 'age_nl2cypher_query']

Optional step: reconstruct-graph

In case the graph related data in docker /app/graphrag-folder/output lost, then the graph can be reconstructed using the data stored in DB, which was saved in step 5.

```
$ docker compose up reconstruct-graph
```

```
[+] Running 2/2
 ✓ Container postgres                Running                                                    0.0s
 ✓ Container reconstruct-graph-app   Created                                                    0.3s
Attaching to reconstruct-graph-app
reconstruct-graph-app  | All files have been reconstructed from Azure SQL to /app/graphrag-folder/restore.
reconstruct-graph-app exited with code 0
```

```
$ docker exec -it postgres bash
```

```
root@fffff:/app/graphrag-folder# cd restore
root@fffff:/app/graphrag-folder/restore# ls
communities.parquet        embeddings.community.full_content.parquet  graph.graphml          text_units.parquet
community_reports.parquet  embeddings.entity.description.parquet      relationships.parquet
context.json               embeddings.text_unit.text.parquet          stats.json
documents.parquet          entities.parquet                           summaries.json
```

Query in docker postges

```
$ docker exec -it postgres bash
```

```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()--() RETURN count(*)
$$) AS (count agtype);
 count
-------
 1392
(1 row)

postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()-[r]->() RETURN count(r)
$$) AS (count agtype);
 count
-------
 696
(1 row)
```

Graph storage

Graph metadata, like nodes, edges stored in Docker postgres

```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()--() RETURN count(*)
$$) AS (count agtype);
 count
-------
 1392
(1 row)

postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()-[r]->() RETURN count(r)
$$) AS (count agtype);
 count
-------
 696
(1 row)
```

Graph input and output stored in DB

Example: DB is Azure hosted PostgreSQL

```
postgres=> \dt
                List of relations
 Schema |       Name        | Type  |   Owner
--------+-------------------+-------+----------
 public | graphrag_inputs   | table | XXXX
 public | graphrag_outputs  | table | XXXX
(2 rows)
```

```
$ docker exec -it postgres bash
```

```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# \dt
                List of relations
   Schema   |   Name    | Type  |  Owner
------------+-----------+-------+----------
 ag_catalog | ag_graph  | table | postgres
 ag_catalog | ag_label  | table | postgres
(2 rows)

postgres=# SELECT * FROM ag_catalog.ag_label;
       name        | graph | id | kind |          relation           |        seq_name
-------------------+-------+----+------+-----------------------------+------------------------
 _ag_label_vertex  | 87167 |  1 | v    | "graphRAG"._ag_label_vertex | _ag_label_vertex_id_seq
 _ag_label_edge    | 87167 |  2 | e    | "graphRAG"._ag_label_edge   | _ag_label_edge_id_seq
 Document          | 87167 |  3 | v    | "graphRAG"."Document"       | Document_id_seq
 Entity            | 87167 |  4 | v    | "graphRAG"."Entity"         | Entity_id_seq
 RELATED_TO        | 87167 |  5 | e    | "graphRAG"."RELATED_TO"     | RELATED_TO_id_seq
(5 rows)


postgres=# SELECT * FROM ag_catalog.ag_graph;
 graphid |   name    | namespace
---------+-----------+------------
   87167 | graphRAG  | "graphRAG"
(1 row)
```

Example: DB is Azure SQL database

Run Cancel Disconnect Change Database:

```
1  SELECT TABLE_SCHEMA, TABLE_NAME
2  FROM INFORMATION_SCHEMA.TABLES
3  WHERE TABLE_TYPE = 'BASE TABLE';
4
```

Results Messages

	TABLE_SCHEMA	TABLE_NAME
1	dbo	graphrag_inputs
2	dbo	graphrag_outputs